Integrated Photonic Chip for Large-Capacity Interrogation of Weak Fiber Bragg Grating Arrays

XU KAICHUAN [1,2,3], YE GENGYUAN [4], WANG FENG [1,2,3*], LI MINGLEI [1,2,3], ZHU CHAOBO [1,2,3], YUAN WEI [1,2,3], CHEN XIANGFEI [1,2,3*]

[1] *Engineering Research Center of Precision Photonics Integration and System Application, Ministry of Education, Nanjing University, Nanjing 210093, China.*
[2] *National Laboratory of Solid State Microstructures, Nanjing University, Nanjing 210093, China.*
[3] *College of Engineering and Applied Sciences, Nanjing University, Nanjing 210093, China.*
[4] *Nanjing Jinling High School, Zhongshan Road 169, Nanjing 210005, China*

**Corresponding author: Feng Wang(wangfeng@nju.edu.cn); Xiangfei Chen(chenxf@nju.edu.cn)*

Received XX Month XXXX; revised XX Month, XXXX; accepted XX Month XXXX; posted XX Month XXXX (Doc. ID XXXXX); published XX Month XXXX

We report a monolithically integrated distributed feedback multi-wavelength laser array (DFB-MLA) system-on-chip (SoC) for large capacity interrogation of weak fiber Bragg grating (WFBG) arrays. This ultra-compact architecture integrates functions of laser generation, power regulation, and high-speed modulation, marking a paradigm shift from traditional discrete systems to chip-scale solutions. Leveraging the reconstruction-equivalent-chirp (REC) technique, the 8×3 laser array achieves a precise channel spacing of 2.5 nm, covering a 60 nm tuning range with high fabrication yield. The SoC combines the laser array using an on-chip three-level Y-branch combiner. The combined light is then injected into the primary optical amplifier (PE-OA) to equalize power of different wavelengths. Then a secondary OA (PG-OA) acts as a high-speed modulator to convert the continuous light into optical pulses. Utilizing a time-wavelength division multiplexing (TWDM) scheme, the system demonstrates the simultaneous interrogation of 100 WFBGs with 0.2 m spatial resolution—significantly advancing the state-of-the-art in integrated demodulation. This work underscores the potential of the photonic integrated chip for real-time monitoring in aerospace, structural health, and energy storage systems.

Keywords: *Monolithic integration, Multi-wavelength laser array (MLA), Optical amplifier (OA), weak fiber Bragg grating (WFBG) array interrogation, Reconstruction-equivalent-chirp technique (REC).*

1. INTRODUCTION

Photonic integration has become a key technology for next-generation optical systems due to its high integration density, low power consumption, and excellent reliability [1, 2]. The rapid development of heterogeneous, hybrid, and monolithic integration has enabled lasers, modulators, and detectors to be integrated on a single chip, significantly improving the compactness and scalability of optical communication and sensing systems.

Fiber Bragg grating (FBG) sensors have been widely applied in structural health monitoring, battery safety monitoring, and intelligent transportation because of their immunity to electromagnetic interference, lightweight structure, and large multiplexing capability [3]. To further increase sensing capacity and spatial resolution, weak FBG (WFBG) arrays have attracted considerable attention. However, most existing interrogation systems rely on discrete broadband sources, external modulators, and optical amplifiers, resulting in bulky architectures with limited integration.

Early WFBG interrogation systems employed amplified spontaneous emission (ASE) sources with time-wavelength division multiplexing (TWDM) scheme, enabling large-capacity sensing but requiring multiple discrete optical components [4]. Frequency-domain mode-locked (FDML) lasers later improved interrogation speed, although their fiber-based tunable filters hinder photonic integration [5]. More recently, distributed feedback multi-wavelength laser arrays (DFB-MLAs) have emerged as promising light sources because of their excellent wavelength stability and parallel output capability [6]. Nevertheless, most reported DFB-MLA-based systems still depend on off-chip amplifiers and modulators, limiting integration density, modulation speed, and power efficiency.

To overcome these limitations, we propose a monolithically integrated DFB-MLA photonic chip based on the reconstruction-equivalent-chirp (REC) technique [7, 8]. REC technique enables sub-nanometer wavelength control using micron-scale lithography, improving fabrication tolerance and wavelength uniformity. The proposed chip integrates 24 DFB lasers in an 8 × 3 array with a 2.5 nm channel spacing and a total wavelength coverage of 60 nm. Two on-chip optical amplifiers (OAs) are incorporated to provide optical amplification and high-speed pulse modulation. By integrating multi-wavelength generation, amplification, and modulation on a single chip, the proposed architecture eliminates external optical amplifiers and modulators, significantly simplifying the interrogation system while improving output capability, modulation speed, and scalability for compact, large-capacity WFBG interrogation.

2. IMPLEMENTATION AND TESTING OF THE PHOTONIC INTEGRATED CHIP

The schematic and microscope image of the proposed photonic integrated chip are shown in Fig. 1(a) and Fig. 1(b), respectively. The cavity lengths of the DFB lasers, the power-equalization optical amplifier (PE-OA), and the pulse-gating optical amplifier (PG-OA) are all designed to be 400 μm. The distributed feedback multi-wavelength laser array (DFB-MLA) adopts an 8×3 architecture integrating 24 laser channels with a channel spacing of 2.5 nm, providing a total wavelength coverage of approximately 60 nm (1515–1575 nm). This wavelength range corresponds to a relatively flat material gain spectrum, thereby reducing output power variations among different wavelength channels.

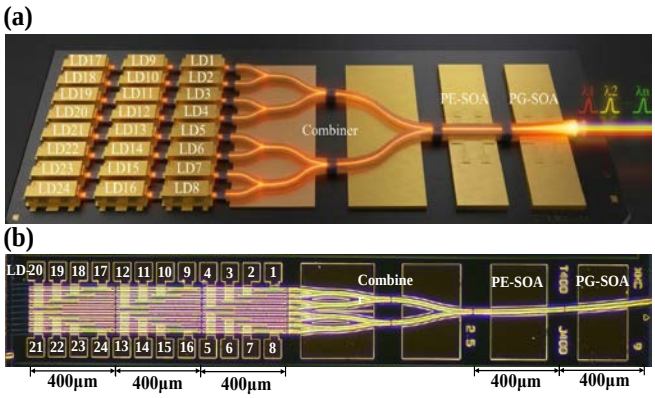


Fig. 1. Monolithically integrated photonic chip. (a) Schematic of the monolithically integrated photonic chip. (b) Microscope image of the fabricated photonic chip.

To evaluate the performance of the proposed chip, the 24-channel DFB-MLA was systematically characterized. As shown in Fig. 2(a), under an injection current of 80 mA per channel at 25 °C, all lasers exhibit stable single-mode operation with side-mode suppression ratios (SMSRs) exceeding 40 dB, while achieving an overall spectral bandwidth of approximately 60 nm. Fig. 2(b) further shows that the wavelength spacing between adjacent channels is approximately 2.5 nm, with wavelength deviations confined within ±0.3 nm. By continuously tuning the injection current from 50 to 180 mA, stable wavelength sweeping is achieved for all channels. The measured tuning ranges, summarized in Table 1, indicate wavelength overlaps exceeding 0.4 nm between adjacent channels, ensuring continuous spectral coverage across the entire sweeping range.

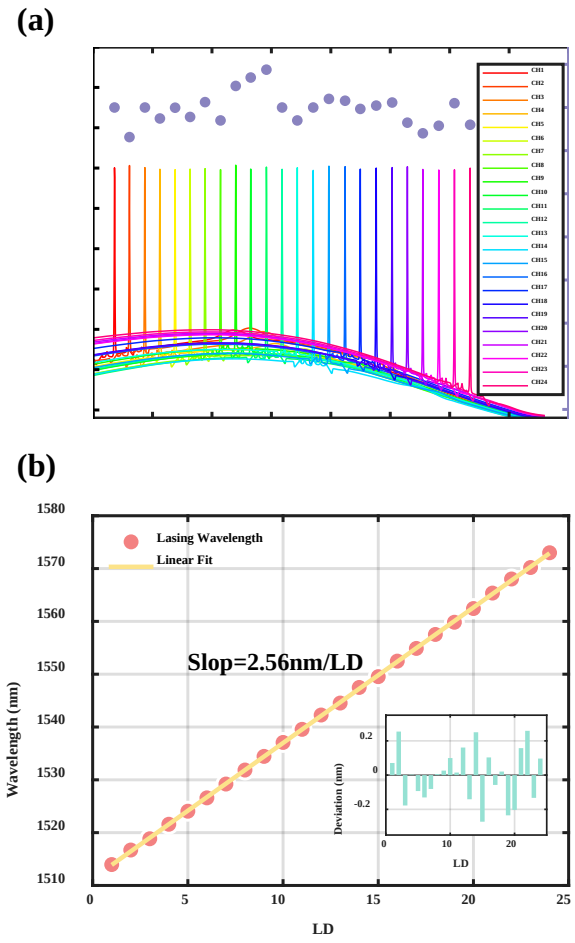


Fig. 2. Spectral characteristics of the 24-channel DFB-MLA. (a) Lasing spectra of the 24-channel DFB-MLA. (b) Channel wavelength spacing and wavelength uniformity.

Table 1. Wavelength sweeping ranges of the 24-Channel DFB lasers.

LD	Wavelength range (nm)	LD	Wavelength range (nm)
1	1513.30–1516.35	13	1540.80–1544.95
2	1515.80–1518.51	14	1544.50–1547.45
3	1517.70–1520.72	15	1547.02–1549.89
4	1520.08–1521.85	16	1549.30–1552.05
5	1521.40–1524.22	17	1551.72–1554.85
6	1523.93–1526.74	18	1554.18–1557.15
7	1526.25–1528.96	19	1556.70–1559.71
8	1528.54–1531.55	20	1559.20–1562.39
9	1531.27–1534.31	21	1561.92–1565.15
10	1533.88–1536.62	22	1564.55–1567.98
11	1536.33–1539.40	23	1567.40–1570.68
12	1538.84–1541.50	24	1570.30–1573.09

Fig. 3 presents the power equalization performance of the PE-OA. Before equalization, the swept spectra exhibit an inter-channel power variation of approximately 10 dB, as shown in Fig. 3(a), mainly due to differences in threshold current, slope efficiency, and gain characteristics among the DFB lasers. By applying pre-calibrated driving currents to the PE-OA through a DAC-based control circuit, active channel-by-channel gain compensation is achieved, reducing the maximum power variation to approximately 0.7 dB, as shown in Fig. 3(b). This significantly improves the spectral flatness and provides a power-balanced broadband source for subsequent high-speed pulse modulation.

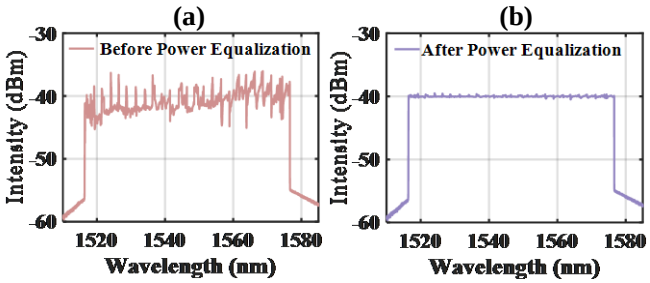


Fig. 3. Power equalization performance of the PE-OA. (a) Swept spectra before power equalization. (b) Swept spectra after power equalization.

The temporal characteristics of the swept source after PG-OA modulation are presented in Fig. 4. Under a modulation condition of a 5 ns pulse width, a 10 ns period, and an 80 mA driving current, the continuous broadband swept light is converted into stable high-speed optical pulse sequences. Across the entire 60 nm operating bandwidth, the pulse amplitudes remain highly uniform, corresponding to a power flatness better than 0.19 dB, while the extinction ratio exceeds 16 dB. The measured rise and fall times are approximately 1.2 ns and 1.0 ns, respectively, demonstrating the excellent nanosecond-scale switching capability and pulse fidelity of the PG-OA, which satisfy the requirements of high-speed time-division multiplexing (TDM) interrogation.

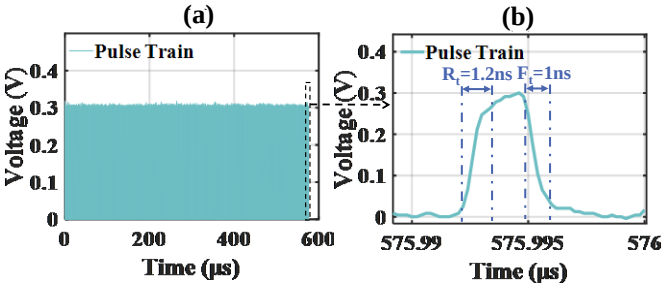


Fig. 4. High-speed pulse modulation performance of the PG-OA. (a) Pulse train. (b) Single-pulse waveform.

To accurately reconstruct the reflection spectra of WFBG arrays, a precise wavelength–time mapping must be established before system operation. Although wavelength overlaps exceeding 0.4 nm are intentionally introduced between adjacent laser channels to ensure continuous spectral coverage, they may lead to wavelength ambiguity during TWDM interrogation if left uncalibrated. Therefore, a wavelength calibration system consisting of a Fabry–Perot (FP) etalon and a tunable narrowband filter is employed to establish an accurate wavelength–time mapping, thereby eliminating demodulation ambiguity caused by spectral overlap between adjacent channels [6].

The experimental results demonstrate that the proposed monolithically integrated photonic chip achieves broadband continuous wavelength sweeping over a 60 nm spectral range, together with excellent wavelength uniformity, inter-channel power equalization, and nanosecond-scale high-speed pulse modulation. These characteristics make it a compact and high-performance swept light source for large-capacity WFBG interrogation systems.

3. EXPERIMENTAL SETUP AND RESULTS

A. Experimental setup

Fig. 5 illustrates the proposed WFBG array interrogation system based on the integrated photonic chip. The pulsed optical output from the photonic chip is divided into two paths by a 10:90 optical coupler (OC). Ninety percent of the optical power is injected into the WFBG array through an optical circulator, and the reflected signals are detected by an avalanche photodiode (APD) and recorded using an oscilloscope (OSC). The remaining 10% of the optical power is monitored by a photodiode (PD). During data processing, the reflected signals are normalized using the monitored optical power to suppress the influence of source power fluctuations and improve the wavelength demodulation accuracy.

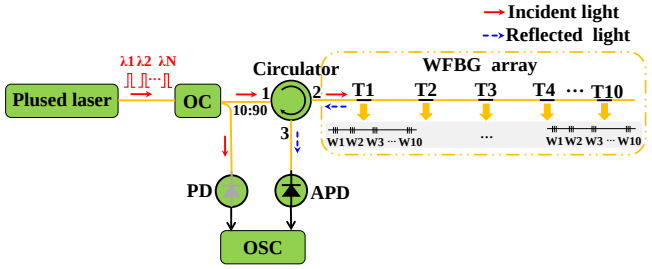


Fig. 5. Experimental setup of the proposed WFBG array interrogation system.

The sensing array consists of 100 WFBGs, each with a reflectivity below 1% and a 3-dB bandwidth of less than 0.3 nm. The array is divided into ten sensing groups (T1–T10), with each group containing ten WFBGs sharing the same Bragg wavelength. The central wavelengths of the ten groups range from 1537.5 nm to 1560 nm with a spacing of 2.5 nm.

Fig. 6 demonstrates the TWDM interrogation performance of the proposed system. First, the WFBGs denoted by (W1)~(T1–T10) were selected to verify the TDM interrogation capability. As shown in Fig. 6(a), when a single-wavelength optical pulse is launched into the sensing array, ten reflected pulses corresponding to the ten sensing

positions are sequentially obtained in the time domain. Fig. 6(b) further presents the reflected signals under multi-wavelength pulse excitation. The PG-OA is driven with a pulse width of 20 ns and a repetition period of 250 ns, while the DFB-MLA performs continuous wavelength sweeping at a tuning rate of 3.47 × 10 ^{4} nm/s. This corresponds to a wavelength interval between adjacent optical pulses is approximately 20 pm. By extracting and normalizing the reflection peaks corresponding to the same WFBG, the reflection spectrum can be reconstructed, enabling accurate determination of the Bragg wavelength.

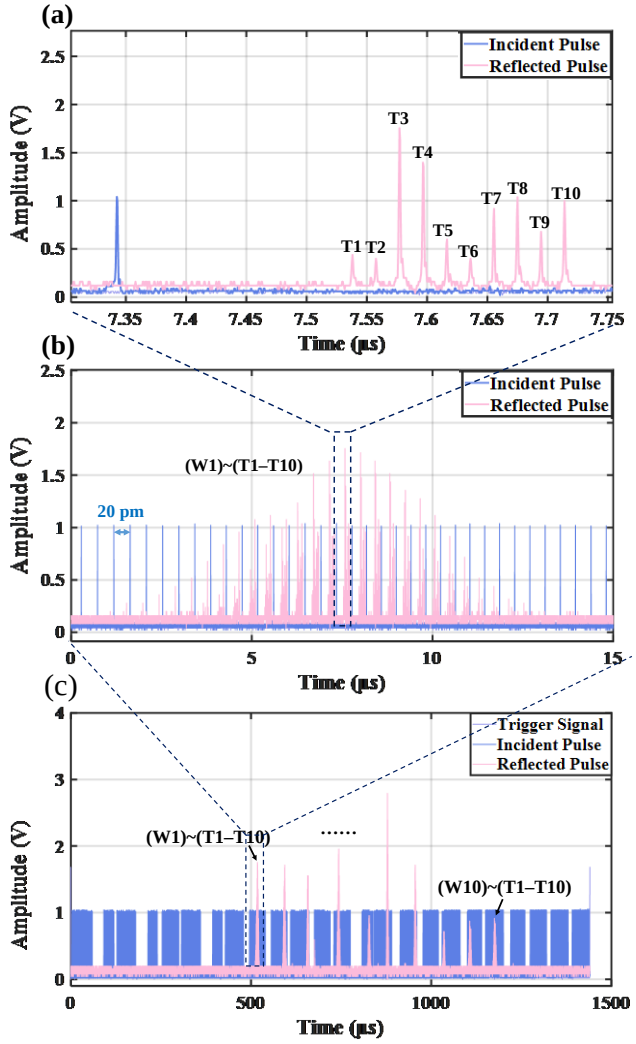


Fig.6. Verification of the proposed TWDM interrogation scheme: (a) TDM interrogation of 10 WFBGs using a single-wavelength pulse; (b) Multi-wavelength pulse response for Bragg spectrum reconstruction; (c) Large-capacity TWDM interrogation of the 100-WFBG array.

Based on the TDM interrogation scheme, wavelength-division multiplexing (WDM) is further introduced to construct a hybrid TWDM system. As shown in Fig. 6(c), during a complete scanning cycle of 1500 μs, the 24 DFB laser channels are sequentially activated, providing continuous wavelength sweeping from 1513.36 nm to 1573.06 nm. Benefiting from the on-chip PE-OA for power equalization and the PG-OA for high-speed pulse modulation, the system generates a power-balanced broadband pulsed light source and successfully interrogates all 100 WFBGs simultaneously. WFBGs with different Bragg wavelengths are distinguished in the wavelength domain, whereas WFBGs sharing the same wavelength are identified according to their temporal positions, demonstrating the large-capacity interrogation capability of the proposed TWDM scheme.

To ensure continuous wavelength sweeping, a certain spectral overlap is intentionally introduced between adjacent laser channels. Therefore, before system operation, wavelength calibration is performed using a Fabry–Perot (FP) etalon and a tunable narrowband filter to establish an accurate wavelength–time mapping. Based on the calibration results, the overlapping wavelength regions between adjacent channels are removed so that each time window corresponds to a unique wavelength. Consequently, the discontinuities observed in the incident pulse sequence in Fig. 6(c) originate from the removal of the overlapping spectral regions, thereby ensuring unambiguous wavelength assignment and accurate TWDM interrogation.

4. CONCLUSION

A monolithically integrated photonic chip for large-capacity WFBG interrogation has been proposed and experimentally demonstrated. The chip integrates a 24-channel DFB multi-wavelength laser array, a PE-OA, and a PG-OA, enabling broadband wavelength generation, active power equalization, and high-speed pulse modulation on a single chip. Experimental results demonstrate a 60 nm wavelength coverage, inter-channel power variation reduced from 10 dB to 0.3 dB, and stable 5 ns optical pulse generation with an extinction ratio above 16 dB. Based on the proposed chip, a TWDM interrogation system successfully interrogates 100 WFBGs. The proposed architecture provides a compact, high-performance, and scalable light-source solution for next-generation large-capacity WFBG sensing systems.

Funding. the Key R&D Program of Jiangsu (BE2023083), the Frontier Technologies R&D Program of Jiangsu (BF2024036) the Frontier Technologies Research Program of Suzhou (SYG202303), the Technology Innovation Foundation of Nanjing University (No. 14913430), the National Natural Science Foundation of China (62404098, 62374092,62371470,62204231,62175100), the Postgraduate Research & Practice Innovation Program of Jiangsu Province (KYCX25_0250) and the Natural Science Foundation of Jiangsu Province of China (No. BK20241196), the 2026 FeiTe Key Research Project, and the 333 High-level Talent Training Project of Jiangsu, Innovation Research Foundation of National University of Defense Technology（Innovation Reserch Foundation of NUDT）

References

1. Yu, C., et al., Advancements in transfer printing techniques and their applications in photonic integrated circuits. Light-Science & Applications, 2025. **14**(1): p. 23.
2. Hu, M., et al., Advances in material, structural, and integration strategies for DFB lasers in photonic sensing platforms. Science China-Information Sciences, 2026. **69**(3): p. 22.
3. Alhussein, A.N.D., et al., Fiber Bragg Grating Sensors: Design, Applications, and Comparison with Other Sensing Technologies. Sensors, 2025. **25**(7): p. 25.
4. Luo, Z.H., et al., A time- and wavelength-division multiplexing sensor network with ultra-weak fiber Bragg gratings. Optics Express, 2013. **21**(19): p. 22799-22807.
5. Liu, Q., et al., High-speed interrogation system of multi-encoding weak FBGs based on FDML wavelength swept laser. Optics and Laser Technology, 2018. **107**: p. 54-58.
6. Xu, K.C., et al., Large-Capacity Interrogation for Ultra-Weak Fiber Bragg Grating Array With a DFB Laser Array Chip. Journal of Lightwave Technology, 2025. **43**(17): p. 8467-8473.
7. Lu, J., et al., Multi-wavelength distributed feedback laser array with very high wavelength-spacing precision. Optics Letters, 2015. **40**(22): p. 5136-5139.
8. Dai, P., et al., Wideband tunable REC-DFB laser array using thin-film heaters on the submount. Chinese Optics Letters, 2023. **21**(1): p. 6.